

Yifan Zhang

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South China University of Technology, Guangzhou, China

RESEARCH PROFILE

Ph.D. student in Software Engineering studying memory enhancement and reliability evaluation for large language model-based agents in scientific research. My work links three areas: multi-source memory enhancement for scientific tasks, reliability evaluation of memory-driven behavior, and construction and validation of domain agents in representative scientific tasks.

PUBLICATIONS

C=CONFERENCE, J=JOURNAL, S=SUBMITTED, A=PREPRINT, T=THESIS, *=EQUAL CONTRIBUTION

- [S.1] **Yifan Zhang**, Ziyang Li, Xiaojing Zhang, Qian Tao (2026). **MemCasa: Can LLM Agents Act on What They Remember?** Manuscript under review.
- [S.2] **Yifan Zhang**, Qian Tao (2026). **Beyond Field Accuracy: Two-Axis Diagnosis of Inverse-PINN Parameter Error.** Manuscript under review.
- [J.1] Sabine Brunswicker, **Yifan Zhang**, Christopher Rashidian, Daniel W. Linna Jr. (2025). **Trust through Words: The Systemize-Empathize-Effect of Language in Task-oriented Conversational Agents.** *Computers in Human Behavior*, 165, 108516.
- [A.1] Shiwen Ni, Guhong Chen, Shuaimin Li, Xuanang Chen, Siyi Li, Bingli Wang, Qiyao Wang, Xingjian Wang, **Yifan Zhang**, Liyang Fan, Chengming Li, Ruifeng Xu, Le Sun, Min Yang (2025). **A Survey on Large Language Model Benchmarks.** *arXiv:2508.15361*.
- [C.1] Tianlu Zheng*, **Yifan Zhang***, Xiang An, Ziyong Feng, Kaicheng Yang, Qichuan Ding (2025). **Gradient-Attention Guided Dual-Masking Synergetic Framework for Robust Text-based Person Retrieval.** *EMNLP 2025*.
- [J.2] **Yifan Zhang**, Christopher Rashidian, Sabine Brunswicker, Dan Whitenack, Daniel W. Linna Jr. (2024). **Empathetic Language in LLMs under Prompt Engineering: A Comparative Study in the Legal Field.** *Procedia Computer Science*, 244, 308–317.
- [C.2] Sabine Brunswicker, **Yifan Zhang**, Christopher Rashidian, Dan W. Linna Jr. (2024). **The Impact of Empathy Display in Language of Conversational AI: A Controlled Experiment with a Legal Chatbot.** *HICSS-57*, 455–466.
- [A.2] Xin Wang*, **Yifan Zhang***, Xiaojing Zhang, Longhui Yu, Xinna Lin, Jindong Jiang, Bin Ma, Kaicheng Yu (2024). **PatentAgent: Intelligent Agent for Automated Pharmaceutical Patent Analysis.** *arXiv:2410.21312*.
- [A.3] Xin Wang, Xiaojing Zhang, **Yifan Zhang**, Xinna Lin, Ruichen Li, Kaicheng Yu (2024). **An Empirical Study on Reconstructing Scientific History to Forecast Future Trends.** OpenReview, 2024.
- [T.1] **Yifan Zhang** (2024). **Advancements in Chest Radiography Pneumonia Classification through Fine-Tuning using MIMIC-CXR-JPG Dataset.** Master's thesis, Northwestern University.

EDUCATION

- **South China University of Technology** Sept 2025 - June 2029 (Expected)
Doctor of Philosophy in Software Engineering Guangzhou, China
 - Advisor: Prof. Qian Tao
 - Dissertation research: memory enhancement and reliability evaluation for LLM-based agents in scientific research
- **Northwestern University** Sept 2022 - April 2024
Master of Science in Computer Science Evanston, US
 - GPA: 3.415/4.00
- **Colby College** May 2022
Bachelor of Arts in Computer Science, Minor in Managerial Economics Waterville, US
 - GPA: 3.14/4.00; Presidential Scholarship recipient

EXPERIENCE

- **South China University of Technology** [🌐] Sept 2025 - Present
Guangzhou, China
PhD Student
 - Conducting doctoral research under Prof. Qian Tao on an evidence–memory–behavior–scientific–outcome pipeline for reliable LLM-based scientific agents.
 - Led two first-author manuscripts on memory-in-action evaluation for LLM agents and diagnostic reliability evaluation for inverse physics-informed neural networks.
 - Developing validation-gated memory methods and planning domain-agent testbeds for representative scientific tasks.
- **DeepGlint** [🌐] March 2025 - Sept 2025
Beijing, China
MLE Intern
 - Co-developed **HMDocRAG-Bench**, a hierarchical benchmark with 6,000 question–answer pairs for evaluating document retrieval-augmented generation systems.
 - Engineered an LLM-based data-generation pipeline spanning six difficulty levels, including multi-hop, multimodal, and multi-page reasoning.
 - Contributed to the MLLM-based data-construction pipeline for the five-million-sample **WebPerson** dataset used in GA-DMS (EMNLP 2025).
- **AutoLab at Westlake University** [🌐] June 2024 - Sept 2025
Hangzhou, China
Visiting Student
 - Conducted research on LLM-based agents, scientific knowledge synthesis, and AI for science.
 - Co-led **PatentAgent**, an agent system for automated pharmaceutical patent analysis, and contributed to research on reconstructing scientific history for trend forecasting.

SELECTED RESEARCH PROJECTS

Agent Memory and Scientific Reliability

- **MemCasa / SmartMem** 2024 - Present
(Manuscript under review)
LLM Agents, Memory Systems, Benchmarking
 - Designed a deterministic memory-in-action benchmark that tests whether remembered information produces correct executable behavior, rather than measuring recall alone.
 - Built a canonical suite of 840 evaluation units across six memory demands with strict and graded scoring and controlled memory policies.
 - On a fixed GPT-4.1-mini suite, structured-memory conditions increased exact success from 0.000 without memory to as high as 0.465.
- **Beyond Field Accuracy / Inverse-PINN Reliability** 2024 - Present
(Manuscript under review)
Physics-Informed Neural Networks, Scientific Reliability, Diagnostics
 - Introduced a two-axis diagnosis that separates finite-sample observation resolution from the learned field’s signed residual preference when explaining inverse-PINN parameter error.
 - Across 240 fresh-noise retrainings, the diagnosed displacement tracked the returned signed log-parameter error at $r = .994$, providing a model for reliability attribution beyond aggregate field accuracy.
- **Procedural Memory Contracts** 2026 - Present
(Ongoing research)
LLM Agents, Procedural Memory, Verification
 - Developing localized, verifier-gated skill updates that attach evidence, scope, preservation constraints, and regression checks to candidate procedural memories.
 - Current pilots expose regression and negative transfer; held-out generalization testing is planned.

Domain Agents for Scientific Tasks

- **PatentAgent** 2024
(arXiv:2410.21312)
LLM Agents, Retrieval-Augmented Generation, Tool Use
 - Co-developed an agent workflow for automated pharmaceutical patent analysis, including evidence retrieval, tool-assisted analysis, and structured reporting.

TEACHING EXPERIENCE

- **Project Manager, Department of Computer Science**

CS396: *Intro to Game Design*

Fall 2023

Northwestern University

- Provided over 6 hours of weekly assistance, including 4 hours dedicated to debugging, resolving student queries, and 1 hour of online support.
- Facilitated student learning in **game physics, Unity, Visual Studio, and C++ programming**.
- Participated in weekly 1-hour TA meetings to review progress, coordinate tasks, and plan examinations.

- **Teaching Assistant, Department of Mathematics**

MA 125: *Single-Variable Calculus*

Spring 2020

Colby College

- Supported grading responsibilities and delivered in-person tutoring sessions to reinforce student understanding of calculus principles.
- Simplified complex calculus concepts using **visual diagrams** and **algebraic formulas**, enhancing comprehension for diverse learning styles.

- **Teaching Assistant, Department of Computer Science**

CS 15X: *Computational Thinking*

Fall 2019

Colby College

- Guided students through fundamental computer science topics, with a focus on **Python programming, syntax, and debugging techniques**.
- Conducted practical sessions to introduce essential debugging tools such as print statements, fostering problem-solving skills.
- Provided tailored support for students with no prior programming experience, making foundational concepts accessible and engaging.

SKILLS

- **Programming Languages:** Python, Julia, R, MATLAB, C/C++, Java, SQL
- **Data Science & Machine Learning:** PyTorch, TensorFlow, scikit-learn, Pandas, NumPy
- **Specialized Areas:** LLM Agents, Agent Memory, Reliability Evaluation, AI for Science, NLP
- **Research Skills:** Experimental Design, Benchmark Construction, Intervention-Based and Diagnostic Evaluation, Scientific Machine Learning, Technical Writing

ADDITIONAL INFORMATION

Languages: Mandarin (Native), English (Fluent), Spanish (Intermediate)

Interests: Run (Half-Marathon), Music (Flute in Jazz Band)